

Supplementary Material

Black Hole and Neutron Star Classification Beyond Known Sources: An RXTE/PCA Evaluation

Youhan Wanigasuriya

This supplement accompanies the integrated manuscript. Its machine-readable evidence index records the original and packaged SHA-256 for every accompanying table and evidence file. Most copies are byte-identical; machine-specific path prefixes are normalized in the remaining copies. Numerical fields are unchanged. The accompanying column dictionary documents table fields and explicitly recorded units. These materials describe completed experiments. They do not contain a newly trained model or a state-matched result.

S1 Source census, observation provenance and cohort definitions

Tables S1 comprise the full reference roster, source registry, reviewed aliases, label-provenance records, selected ObsIDs, exclusions and cohort census. The catalogue and central identity evidence is given by ¹⁻⁴. The reference roster includes rejected and unresolved entries; it is not the final admitted sample. The canonical source identifier is the grouping key. A different catalogue string does not create a new physical system.

The primary reconstruction required four usable observations per source, whereas the later independently processed cohorts required two. This corrects a generalization in the protected historical Methods draft, not a dataset or protocol. The primary cohort has 687 observations (122 BH and 565 NS) from 29 sources (7 BH and 22 NS). The independently processed original-source cohort has 207 observations; the expanded sample has 213, with 9 BH and 22 NS systems. These counts are copied in `S1_cohort_census.csv`.

The repository distributes observation inventories, selections and exclusions under `data/manifests/`, with download URLs and raw checksums in `data/provenance/rxte_download_manifest.csv`. Full per-observation reduction lineage, including independent reduction processing. `json` records, remains in the original research workspace and is not distributed here. The root `RELEASE_MANIFEST.csv` indexes the current package. Raw and native-array products are not included; archive access and distribution limits are described in repository-root `data/README.md`, `docs/data_provenance.md` and the main paper's Data and software availability section. A local dossier is evidence documentation, not a claim that every source has an independently complete physical-state history.

S2 Complete performance, sensitivity and fingerprint results

Table S2 primary metrics includes prior, logistic and forest

results for each split, with AUROC, BA, MCC, F1, precision, recall, average precision, Brier score and accuracy plus available intervals. “PR_AUC” is average precision. The prior uses a fixed NS probability and a tie-to-NS threshold; it is not the fold-dependent empirical prevalence. All headline rows represent physical sources, not observations.

Paired split differences resample identical sources across observation-wise and grouped predictions. The robustness table retains observation-weighted training, training-only principal-component sensitivity, spectra-plus-errors, removal of the most observed source, the high-rate restriction, alternative grouped seeds and fixed logistic settings. These are sensitivity analyses, not multiple independent validations or a model-selection tournament. The learning-curve table is an exploratory fixed-model diagnostic; increasing training-source count in this small sample does not establish an asymptotic population learning curve.

The randomization tables retain every tested label permutation, including low and high grouped outcomes. Label counts are preserved at the source level. The identification-prediction file permits direct calculation of the mean per-source identity accuracy without treating prolific sources as independent replicates. It is not a BH/NS confusion table.

S3 Source-level predictions and fit audit

Primary LOSO scores and the complete grouped/LOSO source summaries for `HEASOFT_VALIDATION`, `HEASOFT_MATCHED`, `EXPANDED_SOURCE` and `PRIMARY_SAME_OBSIDS` are supplied for both classifiers. Tables preserve all sources, including errors and scores near the fixed threshold. Their `p_NS` field is mean-log-odds pooled; `mean_probability` is a different arithmetic summary and must not be substituted when recalculating headline metrics.

Complete observation-level run records and `fit_audits.json`, including outer membership, inner membership, fitted feature columns and scaler/model provenance, remain in the original research workspace and are not distributed here. The current repository includes primary out-of-fold predictions and `data/manifests/primary_split_definitions.json`, alongside the source summaries described above. Per-source probability ranges summarize variation between observations, not confidence intervals for an underlying calibrated class probability. The main Figure 5 intentionally plots no invented source-level error bars.

S4 Independent processing, detector support and confounders

Complete validation metrics transcribes every available metric and interval from the immutable validation and diagnostic JSONs. Its rows retain the original run/tag and source hash. Paired processing effects subtract stored original-product metrics from stored HEASoft metrics on identical ObsIDs and attach the existing paired intervals. No new bootstrap is run. The full JSON structures are also supplied, with any machine-specific path prefixes normalized, so extraction can be audited.

Metadata-only, normalized-shape and background-fraction cuts are distinct controls. Common support uses gain epoch and selected-PCU configuration strata plus exposure, rate and background bounds; the residual imbalance table documents why the word “matched” does not mean exact distributional equality. Epoch and PCU composition tables must be read with the source counts: a source can span epochs, so counts across epochs are not independent systems to be summed.

The epoch table preserves infeasible epochs alongside the feasible case. That case has only two BH and two NS test systems: AUROC is unity for both models, but BA is 0.5 and MCC is zero, with all sources assigned NS. Its degenerate conditional AUROC interval is not an estimate of negligible uncertainty in future sources. Failure annotations and coefficient-stability outputs are descriptive; they do not identify a physical cause of an error.

S5 State evidence, extension and proxies

The original and merged state catalogues are supplied separately. The merged file preserves original fields and extension provenance. DIRECT native-state evidence is not automatically DIRECT common-regime evidence. Only full-observation support and an explicit source-specific spectral interpretation admit a shared regime; provisional, unresolved, conflicting and partial-overlap evidence remains outside the gate.

Twenty copied target dossiers record the bounded NS extension. For IGR J00291+5934, exact early-pointing assignments and spectral support admit the designated hard observations^{5,6}. For SAX J1808.4–3658, native island evidence is supplemented by the source-specific supported interval for the relevant outburst^{7,8}. Other native-only assignments are retained without an automatic cross-class conversion. Burst occurrences alone do not label persistent spectral state.

The final regime table reports 6 BH/4 NS systems in the hard-dominated regime and 2 BH/1 NS in the soft-dominated regime. The closure record retains the five-systems-per-class requirement, infeasible branch, and zero model runs. There is no state-matched metric table with inferred or placeholder performance values.

Proxy metrics and paired proxy-minus-full differences describe the separate three-coordinate diagnostic on the expanded sample. The proxies are fixed detector-space log hardness ratios and log intensity. They do not supply native states. Zero-crossing paired AUROC intervals do not estab-

lish equivalence, complete mediation by state, or absence of information in the finer spectrum.

Native-state and within-source summaries are explicitly descriptive and have source-dependent coverage. The retained reliability figure (Fig. S1) and reliability data do not establish deployment calibration. The same physical system may contribute to multiple native-state categories; those category counts are not interchangeable with independent-source totals.

S6 Software, integrity and reproducibility

Frozen primary, HEASoft, state and extension plans are copied with recorded environment snapshots. The environment correction distinguishes calibration files from directory entries while preserving the original snapshot. The manuscript uses local stage-wise protocol freezing, not external registration of the entire project.

The distributed `tests/` suite and `scripts/verify_release.py` check implementation and release-integrity contracts; repository-root `docs/reproducibility.md` documents the commands and excluded full-workspace tests. Such checks verify implementation properties and fixtures, not a new external validation. Historical manuscript-assembly test logs and the numerical ledger are not part of the distributed release. No scientific fit was executed during assembly. Scientific outcomes come from frozen tables/JSONs, method definitions from frozen protocols/code, and identifiers/citation years are not treated as estimated measurements.

The project reproduction commands distinguish safe verification and manuscript rendering from optional isolated historical reruns. Some historical “audit” scripts write reports, so their names alone do not establish read-only behaviour. Clean-host raw reduction and archival deposition remain release tasks rather than completed claims. Central weaknesses – small source counts, state infeasibility, conditional uncertainty and mission specificity – are stated in the main paper as well as here.

Guide to the accompanying machine-readable material

The table files and evidence dossiers are supplied alongside this PDF in the accompanying table directory. Filenames beginning S1–S6 correspond to Sections S1–S6 above. Table sets retain the numbering of the reviewed manuscript; Table S1 is an extract of set S1, and the reliability figure is Fig. S1. No fitting or resampling was performed to prepare this material. Distributed acquisition inventories and checksums are described in S1. Complete observation-level fit audits and pre-release software history are not distributed in the current release; their local existence does not establish public availability.

File groups and interpretation

S1: Reference roster, aliases, label evidence, selected and excluded observations, and cohort counts.

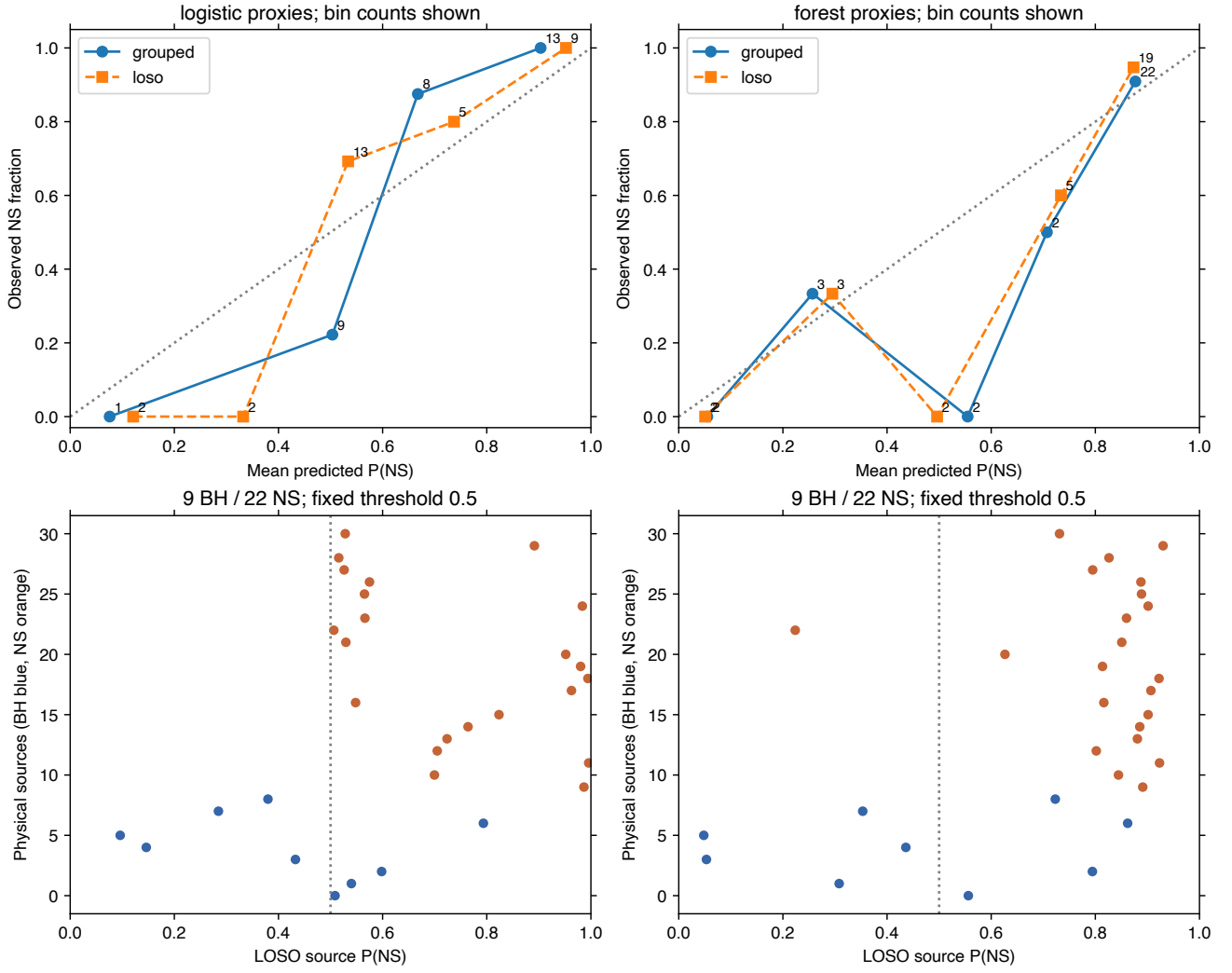


Figure S1: Retained source-level reliability diagnostic for the three-coordinate hardness–intensity proxy models, distinct from the full-spectrum models in main Figure 5. The plotted points and reference lines are unchanged from the frozen figure. These descriptive diagnostics do not establish deployment calibration.

Alt text: Upper panels compare mean predicted neutron-star scores with observed neutron-star fractions and diagonal calibration references for logistic regression and the forest. Lower panels show source-score distributions by true class and the fixed classification threshold.

Table S1: Complete expanded-cohort source roster from the frozen logistic LOSO summary (table set S1). The observation counts refer to the expanded independently processed sample.

Source identifier	Source name	Class	Observations
1a_1744_361	1A 1744-361	NS	8
4u_0614_09	4U 0614+09	NS	8
4u_1254_69	4U 1254-69	NS	8
4u_1543_475	4U 1543-475	BH	4
4u_1608_522	4U 1608-522	NS	8
4u_1636_536	4U 1636-536	NS	8
4u_1702_429	4U 1702-429	NS	8
4u_1705_44	4U 1705-44	NS	8
4u_1728_34	4U 1728-34	NS	8
4u_1735_444	4U 1735-444	NS	8
4u_1746_37	4U 1746-37	NS	8
aql_x_1	Aql X-1	NS	8
cyg_x_2	Cyg X-2	NS	7
gro_j1655_40	GRO J1655-40	BH	2
grs_1915_105	GRS 1915+105	BH	4
gs_1354_64	GS 1354-64	BH	7
gx_339_4	GX 339-4	BH	7
hete_j1900_1_2455	HETE J1900.1-2455	NS	8
igr_j00291_5934	IGR J00291+5934	NS	4
ks_1731_260	KS 1731-260	NS	8
mx_b_1658_298	MXB 1658-298	NS	8
sax_j1750_8_2900	SAX J1750.8-2900	NS	8
sax_j1806_5_2215	SAX J1806.5-2215	NS	8
sax_j1808_4_3658	SAX J1808.4-3658	NS	7
sax_j1810_8_2609	SAX J1810.8-2609	NS	7
ser_x_1	Ser X-1	NS	8
slx_1735_269	SLX 1735-269	NS	8
v4641_sgr	V4641 Sgr	BH	5
xte_j1118_480	XTE J1118+480	BH	7
xte_j1550_564	XTE J1550-564	BH	3
xte_j1859_226	XTE J1859+226	BH	5

S2: Complete primary metrics, paired split effects, sensitivity results, learning curves, randomization and identification predictions.

S3: Complete per-source primary LOSO and independent-processing grouped/LOSO summaries for both classifiers.

S4: Independent-processing metrics, paired effects, detector/epoch composition, residual imbalance, randomization and diagnostics.

S5: Original and merged state catalogues, all target dossiers, closure, feasibility, proxies, paired effects and reliability data.

S6: Frozen protocols, mapping/search plans and recorded software/processing environments.

Column documentation: The accompanying column dictionary documents table fields and explicitly recorded units. The `tables/evidence_index.csv` records original and packaged hashes. Original and packaged hash differences denote path-prefix normalization only. Full-precision numerical strings, row order and scientific annotations are preserved. Any future archival deposit will be identified separately.

References

1. J. M. Corral-Santana, J. Casares, T. Muñoz-Darias, F. E. Bauer, I. G. Martínez-Pais, and D. M. Russell. BlackCAT: A catalogue of stellar-mass black holes in X-ray transients. *A&A*, 587:A61, 2016. doi: 10.1051/0004-6361/201527130. URL <https://doi.org/10.1051/0004-6361/201527130>.
2. Duncan K. Galloway, J. J. M. in 't Zand, Jérôme Chenevez, Hauke Wörmel, Laurens Keek, Laura Ootes, Anna L. Watts, Luis Gisler, Celia Sanchez-Fernandez, and Erik Kuulkers. The Multi-INstrument Burst ARchive (MINBAR). *ApJS*, 249(2):32, 2020. doi: 10.3847/1538-4365/ab9f2e. URL <https://doi.org/10.3847/1538-4365/ab9f2e>.
3. Jerome A. Orosz, Erik Kuulkers, Michiel van der Klis, Jeffrey E. McClintock, Michael R. Garcia, Paul J. Callanan, Charles D. Bailyn, Raj K. Jain, and Ronald A. Remillard. A Black Hole in the Superluminal Source SAX J1819.3-2525 (V4641 Sgr). *ApJ*, 555(1):489–503, 2001. doi: 10.1086/321442. URL <https://doi.org/10.1086/321442>.
4. A. Sanna, E. Bozzo, A. Papitto, A. Riggio, C. Ferrigno, T. Di Salvo, R. Iaria, S. M. Mazzola, N. D’Amico, and L. Burderi. XMM-Newton detection of the 2.1 ms coherent pulsations from IGR J17379–3747. *A&A*, 616:L17, 2018. doi: 10.1051/0004-6361/201833205. URL <https://doi.org/10.1051/0004-6361/201833205>.
5. Manuel Linares, Michiel van der Klis, and Rudy Wijnands. An accreting millisecond pulsar with black hole-like X-ray variability: IGR J00291+5934. *ApJ*, 660(1):595–604, 2007. doi: 10.1086/512666. URL <https://doi.org/10.1086/512666>.
6. M. Falanga, L. Kuiper, J. Poutanen, E. W. Bonning, W. Hermsen, T. Di Salvo, P. Goldoni, A. Goldwurm, S. E. Shaw, and L. Stella. INTEGRAL and RXTE observations of accreting millisecond pulsar IGR J00291+5934 in outburst. *A&A*, 444(1):15–24, 2005. doi: 10.1051/0004-6361:20053472. URL <https://doi.org/10.1051/0004-6361:20053472>.
7. Peter Bult and Michiel van der Klis. The aperiodic X-ray variability of the accreting millisecond pulsar SAX J1808.4-3658. *ApJ*, 806(1):90, 2015. doi: 10.1088/0004-637X/806/1/90. URL <https://doi.org/10.1088/0004-637X/806/1/90>.
8. Askar Ibragimov and Juri Poutanen. Accreting millisecond pulsar SAX J1808.4-3658 during its 2002 outburst: evidence for a receding disc. *MNRAS*, 400(1):492–508, 2009. doi: 10.1111/j.1365-2966.2009.15477.x. URL <https://doi.org/10.1111/j.1365-2966.2009.15477.x>.